INFORMATION TO KEEP WITH YOU

My complete treatment handbook

TCHP · subcutaneous trastuzumab

Understand your treatment. Know what to expect. Keep your plan to hand.

Displayed calendar
21 days
See the active treatment phase

Treatment stages
2
Explained in the treatment reference

Regimen source
NCCP 00731
Version 2

Six 21-day cycles before surgery: intravenous (IV) pertuzumab, fixed-dose trastuzumab under the skin, IV docetaxel and IV carboplatin. Surgery and pathology guide the postoperative HER2 plan. This is not the combined Phesgo injection.

Treatment stages

01 **Current combination**
21-day cycle

02 **Next treatment decision**
Individual plan

Your team confirms the active stage and dates. The full treatment guide explains each step.

Inside your handbook

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My name _____

Hospital / oncology team _____

Use the urgent instructions and contact numbers in your treatment guide. Your team must replace example numbers before patient use. Your individual treatment plan takes precedence.

CHAPTER 01

Understanding your treatment

IN THIS CHAPTER

Why this treatment is offered, what its medicines do, and how the agreed plan fits your care.

When to get help

Emergency: call 112 / 999

Severe breathing difficulty, collapse, seizure or severe chest pain: call emergency services. Do not wait for an online answer.

Contact your oncology service immediately

Temperature 37.5°C or higher, 36°C or lower, shivering or suddenly feeling unwell — even with a normal temperature. Report new breathlessness, chest symptoms, severe diarrhoea, dehydration, jaundice, unusual bleeding or sudden illness promptly. Do not wait for the next planned test.

Follow the urgent contact instructions from your treating hospital. If you cannot reach the team, seek urgent medical assessment. Outside clinic hours, use your instructed out-of-hours pathway or the emergency department.

This page does not diagnose symptoms or decide whether it is safe to stay at home. You do not need to know which medicine caused a symptom before calling.

What each medicine does

Treat HER2-positive early breast cancer as part of a plan intended to reduce the risk of it returning.

TAXANE CHEMOTHERAPY

Docetaxel



Disrupts the cell’s division machinery

Docetaxel stabilises tiny structures called microtubules. Cells need these to separate their chromosomes during division; disrupting the process can stop cancer cells multiplying. Healthy dividing cells are also affected.

Microtubules stabilised → Division disrupted → Cell growth affected

What to watch for

Take prescribed steroid premedication. Report fever, lasting numbness, swelling, sore nails or infusion symptoms.

PLATINUM CHEMOTHERAPY

Carboplatin



Damages deoxyribonucleic acid (the cell's genetic material) (DNA) inside cells

Carboplatin creates links in DNA, making it harder for cells to copy and repair their genetic instructions. Its dose depends on kidney function as well as the intended regimen.

Platinum enters cell → DNA links form → Copying is disrupted

What to watch for

Keep blood-test appointments. Fever, unusual bleeding, repeated vomiting or infusion symptoms need prompt advice.

HER2-TARGETED ANTIBODY

Trastuzumab



Targets HER2 growth signals

Trastuzumab binds to the HER2 protein on cells and helps reduce HER2-driven growth and recruit immune activity. HER2 testing identifies the cancers for which this approach is used.

HER2 on cell surface → Antibody binds → Growth signalling reduced

What to watch for

Keep echo appointments throughout HER2 treatment. Report breathlessness, swelling, palpitations or a reaction; chemotherapy side effects are tracked separately.

HER2-TARGETED ANTIBODY

Pertuzumab



Blocks HER2 pairing with other receptors

Pertuzumab binds a different area of HER2 from trastuzumab. It limits receptor pairing, so the two antibodies act together at different points on the same growth pathway.

HER2 pairing → Pertuzumab blocks pairing → Growth signals reduced

What to watch for

Report diarrhoea early, especially with poor fluid intake. Keep cardiac checks and report infusion symptoms immediately.

The drawings are simplified explanations of drug action, not anatomical pictures or a prediction of response.

Why this treatment?

This is a TCHP pathway before surgery for higher-risk HER2-positive breast cancer. Trastuzumab is a separate injection under the skin, while pertuzumab and chemotherapy are given by intravenous (IV). It is not Phesgo. Cancer stage, fitness, heart and kidney health, route preference and local availability inform the plan; surgery findings guide later treatment.

What could change the choice?

Ask which cancer features, earlier treatments and health factors led to this recommendation, which alternatives were considered and when the decision will be reviewed.

Your treatment and the next step

Schedule

Six 21-day cycles before surgery: IV pertuzumab, fixed-dose trastuzumab under the skin, IV docetaxel and IV carboplatin. Surgery and pathology guide the postoperative HER2 plan. This is not the combined Phesgo injection.

After this phase

After surgery, the pathology helps select the next HER2 medicine. The plan usually aims to complete about one year of HER2 treatment from the first dose. Residual invasive disease may change the treatment. Blood tests and supportive medicines must follow the selected postoperative regimen.

CHAPTER 02

Your schedule, tests and appointments

IN THIS CHAPTER

Your treatment stages, checks and practical preparations, from before treatment through follow-up.

The phases and how they connect

Treatment stages

01 **Current combination**
21-day cycle

02 **Next treatment decision**
Individual plan

Your team confirms the active stage and dates. The full treatment guide explains each step.

This shows treatment timing, not guaranteed symptom-free days. Your team confirms the active phase and dates.

21-DAY CYCLE
Current combination

21-day cycle

Six 21-day cycles before surgery: intravenous (IV) pertuzumab, fixed-dose trastuzumab under the skin, IV docetaxel and IV carboplatin. Surgery and pathology guide the postoperative HER2 plan. This is not the combined Phesgo injection.

INDIVIDUAL PLAN
Next treatment decision

Individual plan

After surgery, the pathology helps select the next HER2 medicine. The plan usually aims to complete about one year of HER2 treatment from the first dose. Residual invasive disease may change the treatment. Blood tests and supportive medicines must follow the selected postoperative regimen.

Relative sequence, not equal lengths or booked dates. Your team confirms actual appointments and the active medicines.

BASELINE → TREATMENT → FOLLOW-UP

Tests and milestones through treatment

Columns show sequence, not equal durations or appointments. A test is only complete when performed and reviewed. Echo timing and cancer-imaging timing are different. Symptoms can bring any check forward.

Check	Before starting	Cycles 1–4	Around cycle 5 / week 12	Surgery / next phase	Ongoing / follow-up
Blood tests	Baseline full blood count (FBC), renal and liver profile	Before each cycle; additional dose checks as prescribed	Continue the required pre-dose checks	Use the selected active regimen's blood-test schedule	Investigate symptoms / persistent abnormalities

Check	Before starting	Cycles 1–4	Around cycle 5 / week 12	Surgery / next phase	Ongoing / follow-up
Echo / cardiac assessment	Baseline cardiac assessment	Earlier for symptoms / concern	Echo during neoadjuvant treatment, around cycle 5	Surgery does not restart the echo clock	Agree follow-up for the active medicine and cardiac risk
Cancer imaging / markers	Baseline disease assessment	Clinical response; imaging if concern	Response review as planned; not tied to echo	Before surgery / pathology	Breast surveillance after early treatment
Surgery / radiotherapy	Agree indication and sequence	No automatic concurrent radiation	Coordinate with both teams	Local treatment as indicated	Review recovery and late effects
Endocrine / bone health	Check hormone receptors / fracture risk	Plan endocrine therapy if indicated	dual-energy X-ray absorptiometry (bone density) (DEXA) when relevant to bone-risk plan	Endocrine timing by the treating team	Individual DEXA / bone follow-up

The team arranges and reviews tests. This guide shows expected timing; it does not record completed appointments. NCCP 00731 v2 · official regimen · ESMO 2024 early breast cancer guideline · treatment and follow-up · ESC 2022 cardio-oncology guideline · cardiac surveillance and risk · NICE NG101 · bone health and follow-up · ESMO metastatic breast cancer guideline · 2026

THE WHOLE TREATMENT JOURNEY

TCHP · subcutaneous trastuzumab: the whole pathway

HER2-positive neoadjuvant breast cancer

Read from left to right. Columns show stages, not equal lengths of time. This is a general plan: no dates or personal information are needed. Your team confirms which steps apply and arranges appointments.

Treatment, tests and reviews across the course

Care / check	Before starting	Cycles 1–4	Around cycle 5 / week 12	Surgery / next phase	Ongoing / follow-up
Cancer medicines NCCP	Confirm diagnosis and exact regimen	See the dose-day map: use the current regimen and active phase	Echo due; treatment review	Surgery/ pathology select postoperative medicine	Review benefit / tolerance and next plan

Care / check	Before starting	Cycles 1–4	Around cycle 5 / week 12	Surgery / next phase	Ongoing / follow-up
Surgery / radiotherapy EARLY	Agree indication and sequence	No automatic concurrent radiation	Coordinate with both teams	Local treatment as indicated	Review recovery and late effects
Blood tests NCCP	Baseline FBC, renal and liver profile	Before each cycle; additional dose checks as prescribed	Continue the required pre-dose checks	Use the selected active regimen's blood-test schedule	Investigate symptoms / persistent abnormalities
Echo / cardiac checks NCCP · ESC	Baseline cardiac assessment	Earlier for symptoms / concern	Echo during neoadjuvant treatment, around cycle 5	Surgery does not restart the echo clock	Agree follow-up for the active medicine and cardiac risk
Scans / tumour markers EARLY	Baseline disease assessment	Clinical response; imaging if concern	Response review as planned; not tied to echo	Before surgery / pathology	Breast surveillance after early treatment
Endocrine / DEXA BONE	Check hormone receptors / fracture risk	Plan endocrine therapy if indicated	DEXA when relevant to bone-risk plan	Endocrine timing by the treating team	Individual DEXA / bone follow-up

What each check is for

1

Blood tests

FBC, kidney and liver profile at baseline and before every 21-day cycle. The team reviews the necessary results before a planned dose.

Check safety before the next dose.

Ask / act: Agree the booking owner, result reviewer and action if overdue.

2

Heart tests

Check heart pumping function before starting HER2 treatment. Repeat echo or multigated acquisition (heart pumping) scan (MUGA) every 12 weeks: around cycle 5, then cycle 9 if 21-day cycles stay on time. Earlier checks are needed for symptoms or abnormal findings. Count elapsed time from the first HER2 dose; changing phase or route does not restart the clock.

Monitor pumping function and relevant rhythm risk.

Ask / **act:** Agree the booking owner, result reviewer and action if overdue.

3



Cancer imaging

Baseline breast and nodal imaging, with staging tests appropriate to the diagnosis. Review response during preoperative treatment and before surgery. After local treatment, arrange breast surveillance; routine computed tomography (CT), positron emission tomography (PET) or tumour-marker testing is not the usual screening plan for someone without symptoms.

Assess response or follow-up needs.

Ask / **act:** Agree the booking owner, result reviewer and action if overdue.

4

Bone health

If the cancer is also hormone-receptor positive, endocrine treatment may form part of the wider plan. Bone-density testing is linked to aromatase inhibitors, ovarian suppression and fracture risk; it is not automatically required because a HER2 medicine is prescribed. Agree baseline DEXA where indicated and individual follow-up.

Assess fracture risk where relevant.

Ask / **act:** Agree the booking owner, result reviewer and action if overdue.

At the next review, ask

- Which treatment phase am I in, and what happens next?
- Which test is expected next, who arranges it, and who explains the result?
- Who should I contact if an expected appointment has not arrived?

This map does not show whether a test has been ordered, booked or completed. Do not wait for a scheduled test if you develop new symptoms.

Your practical treatment plan



STEP 1

Before the first visit

Confirm blood tests, heart assessment, active medicines, route and the consent discussion.



STEP 2

Plan venous access

A cannula, peripherally inserted central catheter (PICC) or implanted port may be used. If a port is chosen, a trained team inserts it in a separate procedure with local anaesthetic and sometimes sedation; the team explains fasting, blood-thinner and wound instructions.

STEP 3

Infusion day

IV pertuzumab and chemotherapy still require venous access. Trastuzumab 600 mg is a separate thigh injection; this is not Phesgo. Staff check the line, give prescribed premedicines and treatment, and observe for reactions. Tell them immediately about burning, pain, rash or breathlessness.

STEP 4

Home and next review

This regimen does not routinely need a take-home infusion pump. Staff disconnect the infusion; a port is de-accessed when appropriate. Keep line-care and next-visit instructions.

Port insertion and removal

The line team explains placement, dressing care and when normal activities can resume. A port is accessed with a special needle during treatment. Removal of the implanted device is a separate planned procedure when no longer needed; do not remove or adjust it yourself.

PICC care and urgent line symptoms

A PICC usually needs regular dressing and flushing appointments even between treatments. Fever, redness, swelling, pain, leakage or damage requires prompt advice. Your hospital supplies the exact care instructions.

Visit time and transport

First visits can take longer because of loading doses, premedicines and observation. Ask the unit about expected time, transport and any escort needed after sedation or line insertion.

Phase-specific visit pattern

Infusion and observation times differ by drug and first versus later doses; the unit confirms the full visit duration, including blood review and preparation.

When scans, heart checks and bloods happen

Agree who books each test, who reviews the result and what happens if a check is overdue.

Blood tests

FBC, kidney and liver profile at baseline and before every 21-day cycle. The team reviews the necessary results before a planned dose.

Heart ultrasound and electrocardiogram (ECG)

Check heart pumping function before starting HER2 treatment. Repeat echo or MUGA every 12 weeks: around cycle 5, then cycle 9 if 21-day cycles stay on time. Earlier checks are needed for symptoms or abnormal findings. Count elapsed time from the first HER2 dose; changing phase or route does not restart the clock.

Scans and tumour markers

Baseline breast and nodal imaging, with staging tests appropriate to the diagnosis. Review response during preoperative treatment and before surgery. After local treatment, arrange breast surveillance; routine CT, PET or tumour-marker testing is not the usual screening plan for someone without symptoms.

Bone health and endocrine treatment

If the cancer is also hormone-receptor positive, endocrine treatment may form part of the wider plan. Bone-density testing is linked to aromatase inhibitors, ovarian suppression and fracture risk; it is not automatically required because a HER2 medicine is prescribed. Agree baseline DEXA where indicated and individual follow-up.

How surgery and radiotherapy fit

Systemic treatment comes before surgery. Lumpectomy or mastectomy is chosen with the surgical team. Radiotherapy, if indicated, follows surgery and recovery; it may overlap postoperative HER2 treatment when both teams agree. It is not automatically concurrent with the chemotherapy phase.

CHAPTER 03

Symptoms and supportive care

IN THIS CHAPTER

Recognising changes, knowing when to call, and using your prescribed supportive-care plan.

Chemotherapy with HER2 treatment

Day 1 → Day 21: a general pattern

This is a visual orientation, not a prediction of your personal cycle. It shows when some effects often emerge and which effects are **not** tied to a particular day.

Chemotherapy with HER2 treatment

Treatment or check	Timing in this view
HER2 medicine · Day 1	 **Day 1** subcutaneous (SC) + IV
Chemotherapy · Day 1	 **Day 1** IV
Administration reaction	 **Days 1–21** Report during or after any visit
Nausea / appetite	 **Days 1–5** May occur early or later
Blood counts / infection	 **Days 1–21** Risk varies; fever needs prompt advice
Fatigue / mouth changes	 **Days 1–21** Variable timing and severity
Nerves / skin / nails	 **Days 1–21** Can build across doses
Heart / breathing symptoms	 **Days 1–21** Report on any day

Planned treatment: Six 21-day cycles before surgery: IV pertuzumab, fixed-dose trastuzumab under the skin, IV docetaxel and IV carboplatin. Surgery and pathology guide the postoperative HER2 plan. This is not the combined Phesgo injection.

Next phase: After surgery, the pathology helps select the next HER2 medicine. The plan usually aims to complete about one year of HER2 treatment from the first dose. Residual invasive disease may change the treatment. Blood tests and supportive medicines must follow the selected postoperative regimen.

Heart checks: Check heart pumping function before starting HER2 treatment. Repeat echo or MUGA every 12 weeks: around cycle 5, then cycle 9 if 21-day cycles stay on time. Earlier checks are needed for symptoms or abnormal findings. Count elapsed time from the first HER2 dose; changing phase or route does not restart the clock.

Why these bands stay broad: Treatment markers show the planned schedule. Symptom bands are broad possibilities, not measured risk curves; serious symptoms can occur at any time. Band length does not show likelihood or severity, and the chart cannot tell you whether it is safe to receive treatment. NCCP 00731 v2 · official regimen ↗

There are no guaranteed "safe days". Fever, shivers, sudden illness, severe breathing changes or chest pain need prompt assessment on any day.

Between-treatment notes

Day 1–21: The current treatment cycle

Six 21-day cycles before surgery: IV pertuzumab, fixed-dose trastuzumab under the skin, IV docetaxel and IV carboplatin. Surgery and pathology guide the postoperative HER2 plan. This is not the combined Phesgo injection. Your team confirms treatment dates and checks. Serious symptoms need advice on any day.

Support around the medicines

1 

The correct supportive prescription

Follow the script for the active medicine, route and treatment day. Do not carry chemotherapy anti-sickness medicines automatically into an antibody-only phase.

2 Keep heart checks on the plan

An echo every 12 weeks is separate from blood tests or cancer imaging. Ask who books and reviews it.

3 Plan for symptoms

Keep your agreed contact numbers and instructions. New symptoms can need assessment before a routine appointment.

Questions worth asking

Who arranges my tests?

Ask the team who books each test, who checks the result and who to contact if an appointment is overdue. This map does not book appointments.

What if I become unwell between visits?

Use the urgent contact plan for sudden illness, fever, chest symptoms, severe diarrhoea or poor fluid intake. Do not wait for a scheduled blood test or scan.

Try explaining the plan back: Which medicines are active now, which days do you receive or take them, and who is booking and reviewing your next tests?

Docetaxel

- Low blood counts, infection and sepsis
- Hair loss, mouth soreness and fatigue
- Numbness / tingling; sometimes lasting
- Fluid retention and nail changes
- Allergic infusion reaction

Carboplatin

- Low blood counts, infection and bleeding
- Nausea, vomiting and fatigue
- Allergic reaction, sometimes after previous exposure
- Kidney or electrolyte changes

Trastuzumab

- Heart pumping-function reduction / heart failure
- Infusion or allergic reaction
- Diarrhoea and fatigue
- Rare serious lung effects

Pertuzumab

- Diarrhoea and dehydration
- Infusion / allergic reaction
- Rash and fatigue

Supportive prescription draft for review. This is a regimen template. It is not an issued patient prescription; individual review and completion are required.

Day 1 · carboplatin combination

Active medicines: Docetaxel + Carboplatin + Trastuzumab + Pertuzumab

Emetogenic risk: High · carboplatin AUC ≥4

Medicine	Dose / route	When
Aprepitant	125 mg orally	Day 1
Ondansetron	16 mg orally	Day 1 Individual dose / QT / hepatic review
Dexamethasone	12 mg orally	Day 1 Reconcile chemotherapy premedication steroids
Aprepitant	80 mg orally daily	Days 2–3
Metoclopramide	10 mg orally up to three times daily	Subsequent days: duration and use to be confirmed Individual contraindications, renal function and duration review
Dexamethasone	8 mg orally daily	Days 2–4 Carboplatin AUC ≥4 antiemetic row; reconcile total steroid exposure

Additional phase-specific support

- Docetaxel: dexamethasone 8 mg orally twice daily for 3 days starting the day before treatment; minimum 3 doses pre-treatment unless contraindicated. If oral doses were missed, any 20 mg intravenous (IV) replacement is a consultant decision.
- Reconcile docetaxel premedication with the antiemetic steroid course and interactions; issue one coherent steroid prescription, not duplicate courses.
- Consider primary granulocyte colony-stimulating factor (white-cell growth support) (G-CSF) using the source and individual risk; product and timing follow the prescribed plan.
- Clinical template for individual review, not an issued prescription. Confirm allergy, pregnancy precautions, interactions and local policy.
- Clinical template for individual review, not an issued prescription. Confirm allergy, pregnancy precautions, interactions and local policy.
- Clinical template for individual review, not an issued prescription. Confirm allergy, pregnancy precautions, interactions and local policy.
- Clinical template for individual review, not an issued prescription. Confirm allergy, pregnancy precautions, interactions and local policy.

Before signing: verify allergies, current medicines, interactions, renal / hepatic function, QT risk, sedation, prior response, total steroid exposure and the rescue-medicine duration. Do not duplicate medicines from another phase.

Individual changes / rescue duration: _____
Prescriber / date: _____ Pharmacy review: _____

Postoperative transition · confirm selected regimen

Active medicines: Selected postoperative medicine

Emetogenic risk: Confirm the selected postoperative regimen

The postoperative medicine is not selected by this template. Open its regimen before issuing supportive medicines.

Additional phase-specific support

- No routine docetaxel or paclitaxel premedication or chemotherapy antiemetic course follows into antibody-only treatment.
- Phesgo or trastuzumab alone generally requires no routine antiemetic prophylaxis. Do not extend that statement automatically to an antibody-drug conjugate.
- Use the separately selected postoperative regimen for its blood tests, premedication, antiemetics and cardiac plan.
- Clinical template for individual review, not an issued prescription. Confirm allergy, pregnancy precautions, interactions and local policy.

Before signing: verify allergies, current medicines, interactions, renal / hepatic function, QT risk, sedation, prior response, total steroid exposure and the rescue-medicine duration. Do not duplicate medicines from another phase.

Individual changes / rescue duration: _____
Prescriber / date: _____ Pharmacy review: _____

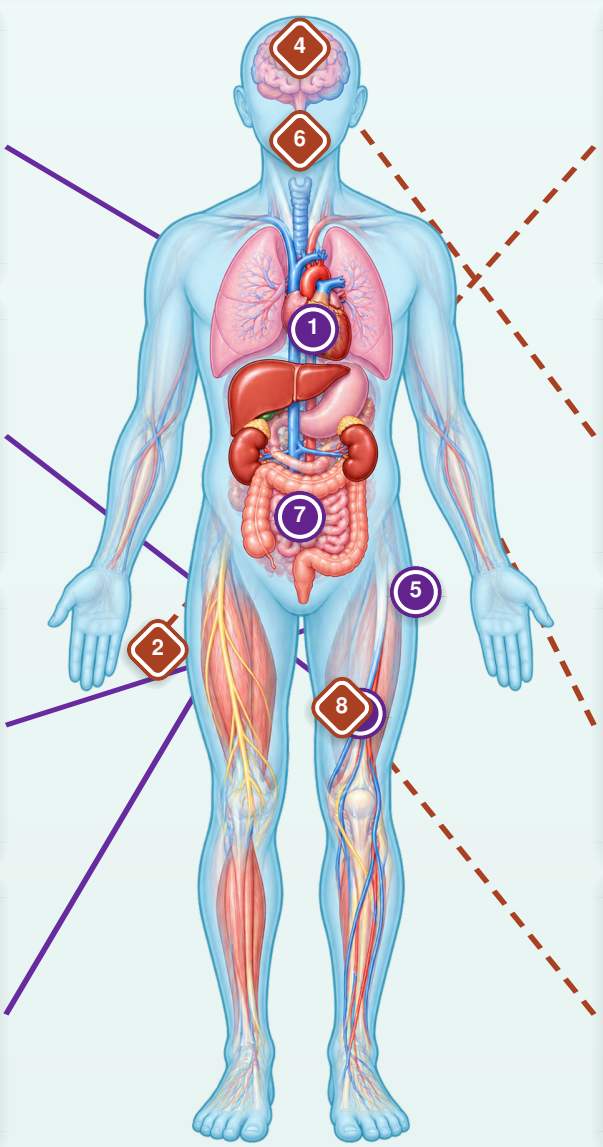
TCHP · subcutaneous trastuzumab

Six 21-day cycles before surgery: intravenous (IV) pertuzumab, fixed-dose trastuzumab under the skin, IV docetaxel and IV carboplatin. Surgery and pathology guide the postoperative HER2 plan. This is not the combined Phesgo injection.

Source reconciled; independent clinical, pharmacy and patient/PPI review pending. Selected effects; not a complete medicine leaflet.

A picture of side effects

Numbered symptom boxes connect to the anatomical illustration. The medicine responsible may be uncertain; report the symptom.



1 Heart

Trastuzumab / pertuzumab

Keep echo appointments; report breathlessness or swelling.

2 Hands / feet

Docetaxel

Report numbness or tingling before the next dose.

3 Blood / infection

Chemotherapy phase

Fever, shivers or suddenly unwell: call immediately.

4 Hair / scalp

Chemotherapy phase

Hair loss is common. Ask about scalp cooling before treatment where suitable.

5 Fluid / skin / nails

Docetaxel

Report swelling, painful skin or nail changes.

6 Mouth

Chemotherapy phase

Painful ulcers or difficulty drinking need advice.

7 Bowel

Pertuzumab / chemotherapy

Report diarrhoea promptly, especially with poor intake.

8 Injection site

subcutaneous (SC) trastuzumab

Report spreading redness, pain or swelling.

Selected effects, not every possible effect or a prediction. Blood-count, skin and line markers are schematic. Number and text carry meaning; colour is decorative.

The map shows selected effects of the medicines named here. Report persistent effects from earlier treatment too.

> Fever, shivers or suddenly unwell: contact oncology immediately. Severe chest pain, breathing difficulty, collapse or seizure: 112 / 999.

CHAPTER 04

Sources and further information

IN THIS CHAPTER

The references behind this information and addresses you can use to find further help.

Use this information alongside the plan agreed with your treatment team. Your team confirms your medicines, doses and appointments.

1. **National Cancer Control Programme (NCCP) 00731 v2 · official regimen**
   https://healthservice.hse.ie/documents/6601/731_v2_TCHSCP.pdf
2. **ESMO 2024 early breast cancer guideline · treatment and follow-up**
   https://www.annalsofoncology.org/article/S0923-7534(23)05104-9/fulltext
3. **ESC 2022 cardio-oncology guideline · cardiac surveillance and risk**
   https://academic.oup.com/eurheartj/article/43/41/4229/6673995
4. **NICE NG101 · bone health and follow-up**
   https://www.nice.org.uk/guidance/ng101/chapter/Recommendations
5. **ESMO metastatic breast cancer guideline · 2026**
   https://www.annalsofoncology.org/article/S0923-7534(26)00902-6/fulltext
6. **EMA Herceptin product information**
   https://www.ema.europa.eu/en/documents/product-information/herceptin-epar-product-information_en.pdf
7. **eviQ · implanted port patient information**
   https://www.eviq.org.au/p/507
8. **eviQ · peripherally inserted central catheter (PICC) patient information**
   https://www.eviq.org.au/p/543
9. **HSE · chemotherapy side effects and urgent advice**
   https://www2.hse.ie/conditions/chemotherapy/side-effects/
10. **NOGG · section 7, aromatase inhibitors and fracture risk**
    https://www.nogg.org.uk/full-guideline/section-7-strategies-management-osteoporosis-and-fracture-risk
11. **NCCP NCIS antiemetics v8 (2025), Table 1**
    https://healthservice.hse.ie/documents/7180/Supportive_Care_Antiemetics_for_inclusion_NCIS_Medical_Oncology.pdf
12. **ASCO · biomarkers in metastatic breast cancer**
    https://ascopubs.org/doi/10.1200/JCO.2015.61.1459
13. **EMA Perjeta product information**
    https://www.ema.europa.eu/en/documents/product-information/perjeta-epar-product-information_en.pdf